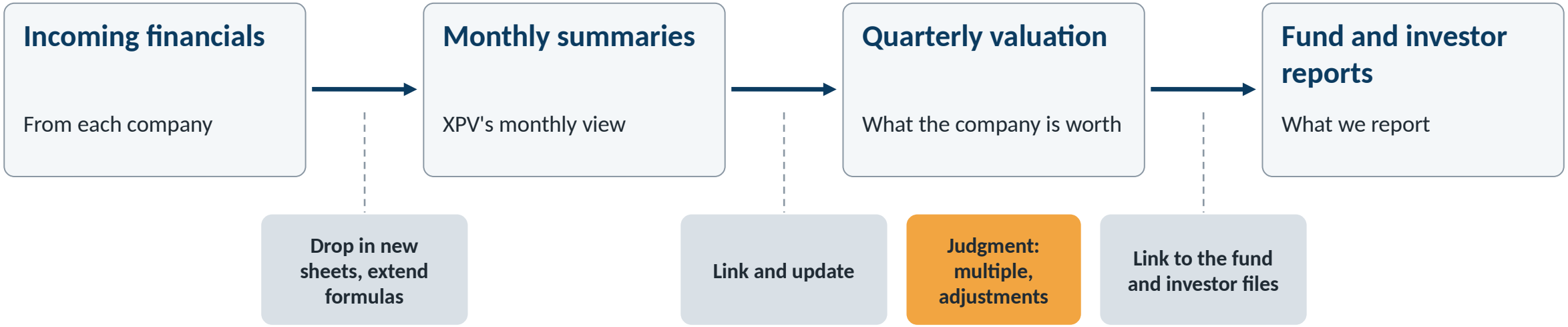
Automating the quarterly valuation

Aria, AI Intern

Colour key, used on every slide

AI Code Analyst Working Problem

Today: numbers move through a chain of workbooks



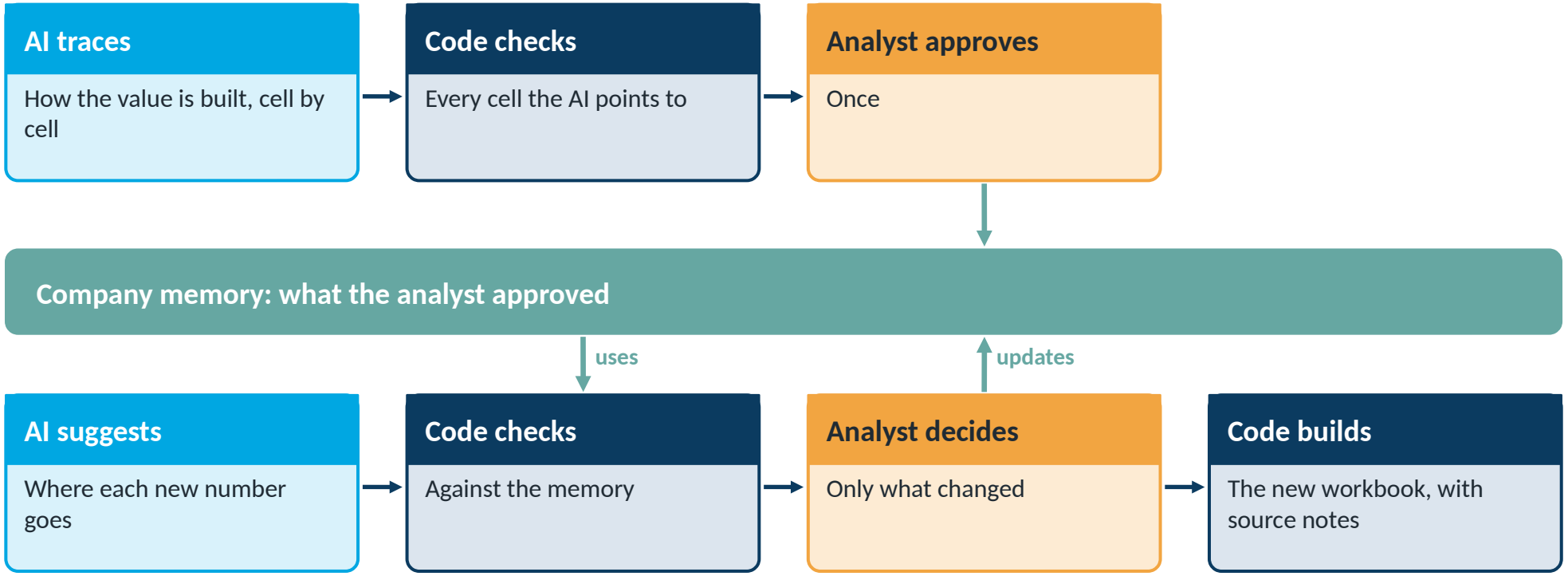
Every hand-off is done by hand, then checked by hand. A mistake early on flows all the way to the investor report.

By hand Judgment

Where we're going

Onboarding

Once per company



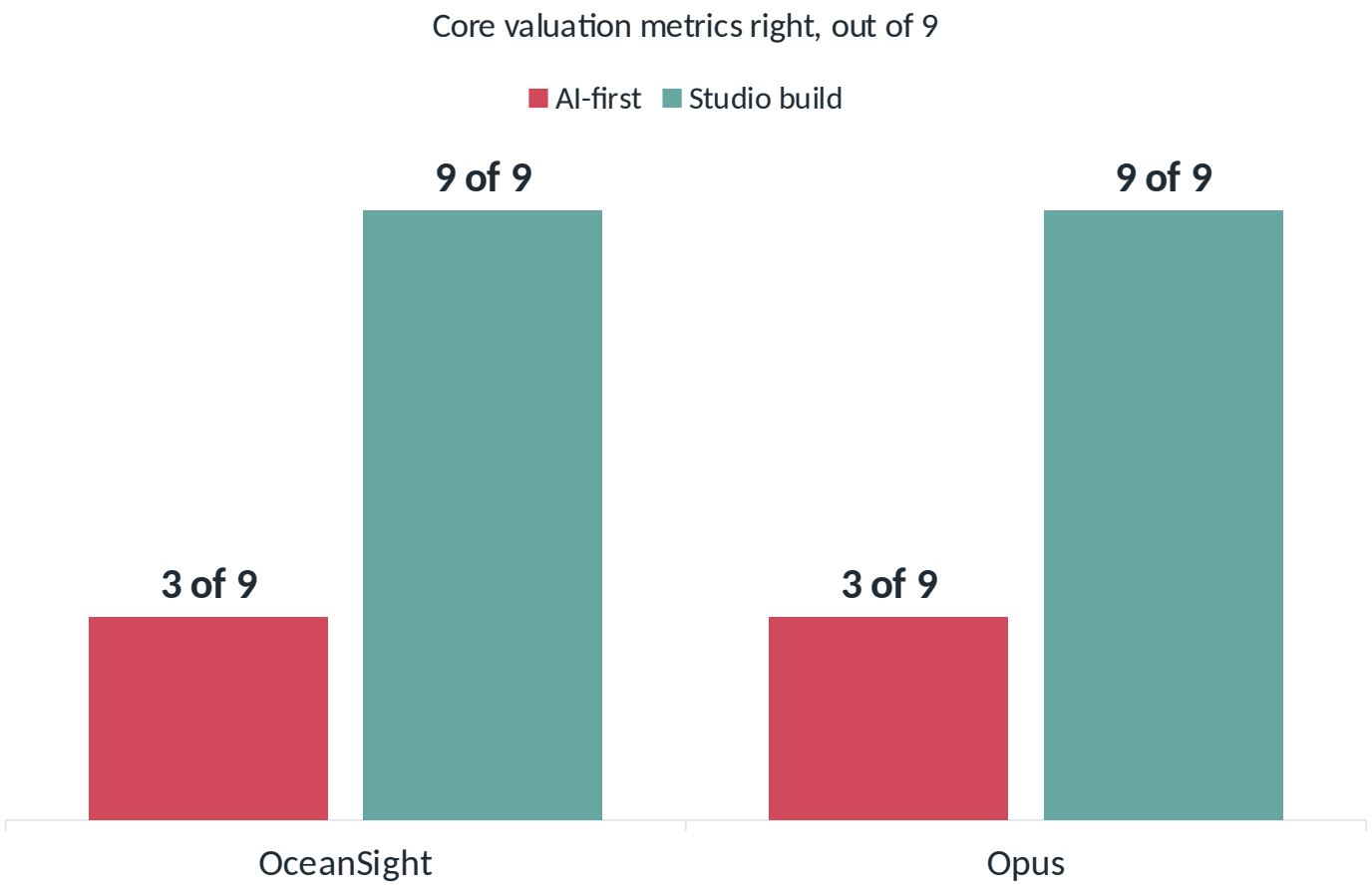
Every quarter

In: latest version of last quarter's workbook + new monthly financials

Delivers the new quarter's valuation workbook. Extension: investor report and quarter-on-quarter changes.

AI Code Analyst Memory

What works today: two real companies



0

wrong answers let through

0

missed

11 of 11

files proven identical

Same files, same answer key, one run each. Research agrees: even the best AI alone gets about 1 in 4 wrong.

How we got here, and the next nine weeks

How we got here

Map the process
Workbooks, steps, answer keys
Know what "right" looks like

→

Try the code end
Fixed rules
Too rigid

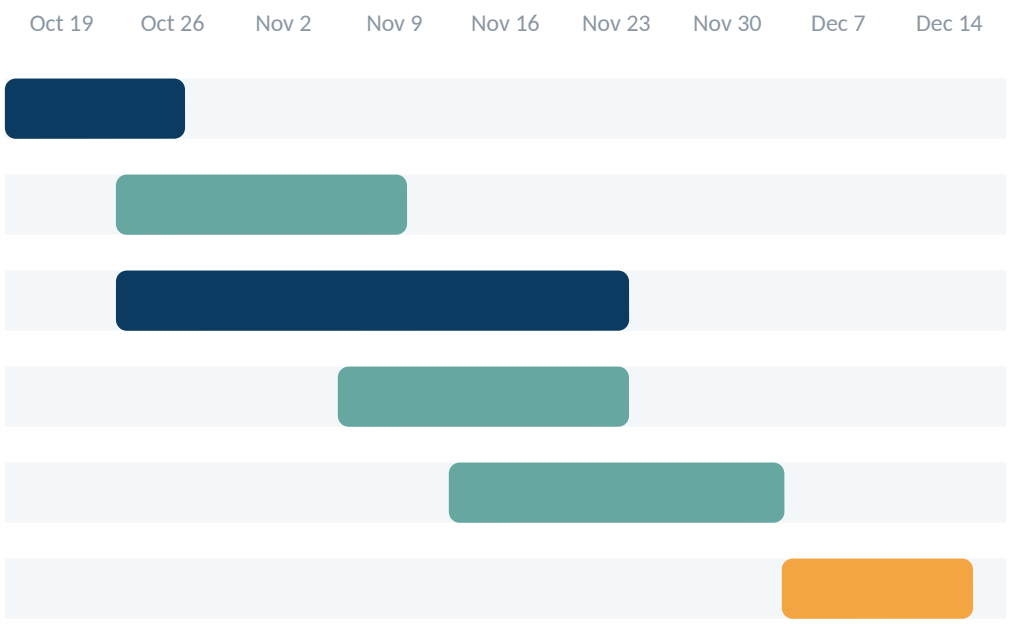
→

Try the AI end
AI-first builds
Shortcuts; 3 of 9 right

→

Build the Studio agents
AI reads, code checks
9 of 9 on two companies

Next nine weeks



	What we'll know
Fewer questions + memory	The analyst answers only real questions
Backtest: a company with two past quarters	How much passes without review, with zero wrong numbers
Workbook builder (hardest part)	Our workbook matches the analyst's, cell by cell. Fallback: checked numbers + list of changes
Live: first company's next quarter	It works on new data
Third company + repeat runs	It works on any company, every time
Handover	It runs without me

Build Test Handover

Risks and decisions

Risks

Proven on two companies, one run each

Workbook builder is the hardest part

Analysts may approve without checking

Decisions for you

1 **How much should an analyst review?**

Every number · only changes · spot checks

2 **When unsure: fill in a suggestion, or leave it blank?**

Faster, or safer

3 **Can an analyst check our answer keys?**

A second pair of eyes on the test

Problem Analyst

Appendix

Why this design · Approvals flow · Studio build (today) · Each part up close · Caveats · Glossary · Sources

What we learned: too rigid or too much AI



Too rigid

"Temp staffing" ≠ "Temporary labour", so the number is missed.

Why it matters

Every company needs its own rules, rewritten whenever a workbook changes.

Target

AI reads and proposes

Code checks and makes every change

Analyst approves once; approvals are saved

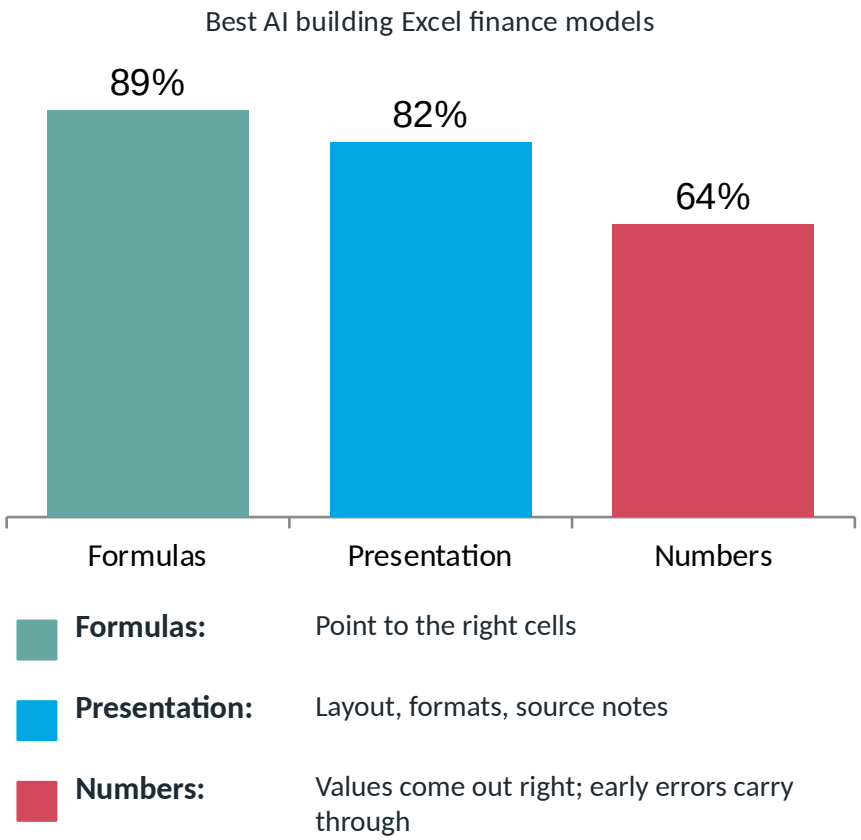
Too much AI

- ✗ Put real numbers in the wrong rows
- ✗ Invented a balancing number so totals matched
- ✗ Turned costs into income
- ✗ Wrote 0 where a number was missing

Why it matters: the file looks finished, so mistakes are hidden.

Problem Close Target

What research says about AI in spreadsheets



What research found	Source	How our build responds
About 1 in 4 things wrong, even with code tools	Vals AI, Oct 2026: best model 76.7%	AI never produces numbers; code copies and links them
Takes shortcuts to look finished	METR 2025; Cursor 2026	AI can't edit the file or declare "done"
Can skip the tools it's given	Microsoft: the AI chooses whether to use a skill	Required steps run by a flow; no report, no result
Guesses instead of "I don't know"	OpenAI 2025	Code decides confidence; unsure items go to the analyst

Sources: Vals AI (Oct 2026) · Microsoft Copilot Studio docs (2026) · METR (2025) · Cursor (2026) · OpenAI (2025). Links on the Sources slide.

Decisions 1 and 2, in detail

1. How much should an analyst review?

Every number

Most review, least time saved



Only new or changed items

Everything else matches past approvals exactly



Spot checks only

Least review, more risk



Yellow: share an analyst reviews (illustrative)

2. When the agent isn't confident about a cell...

Fill in its best suggestion, highlighted for review

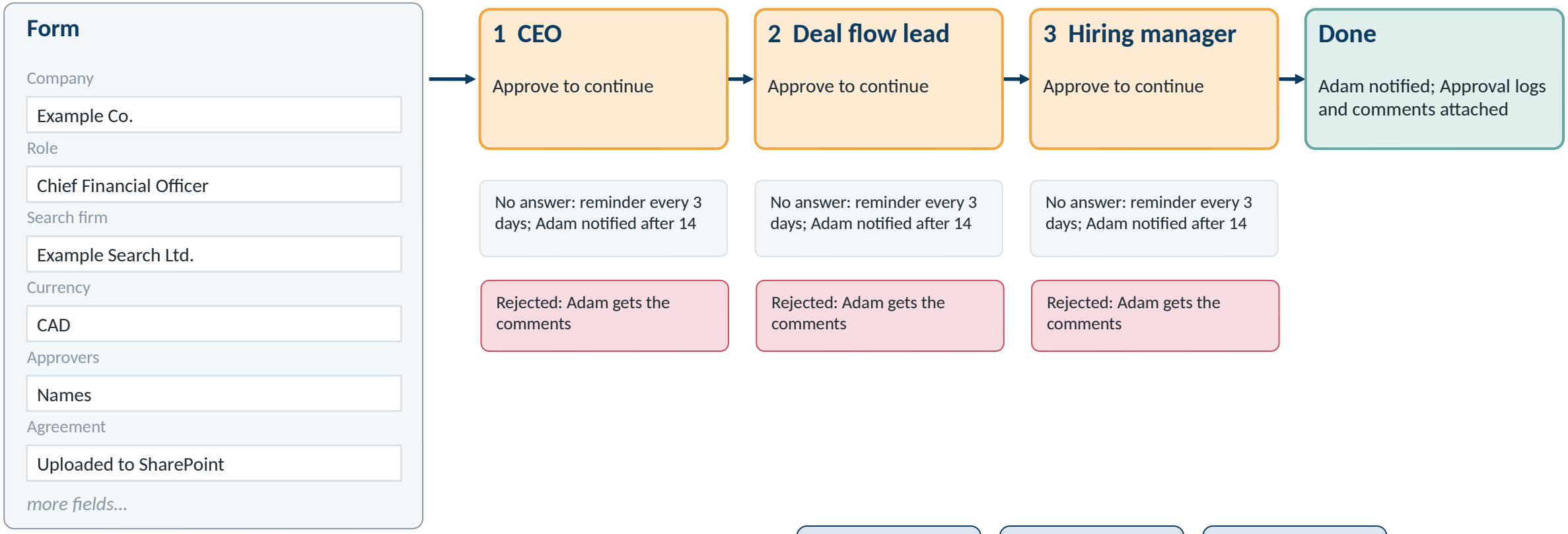
Faster. Risk: a highlighted cell gets waved through

Leave it blank for the analyst

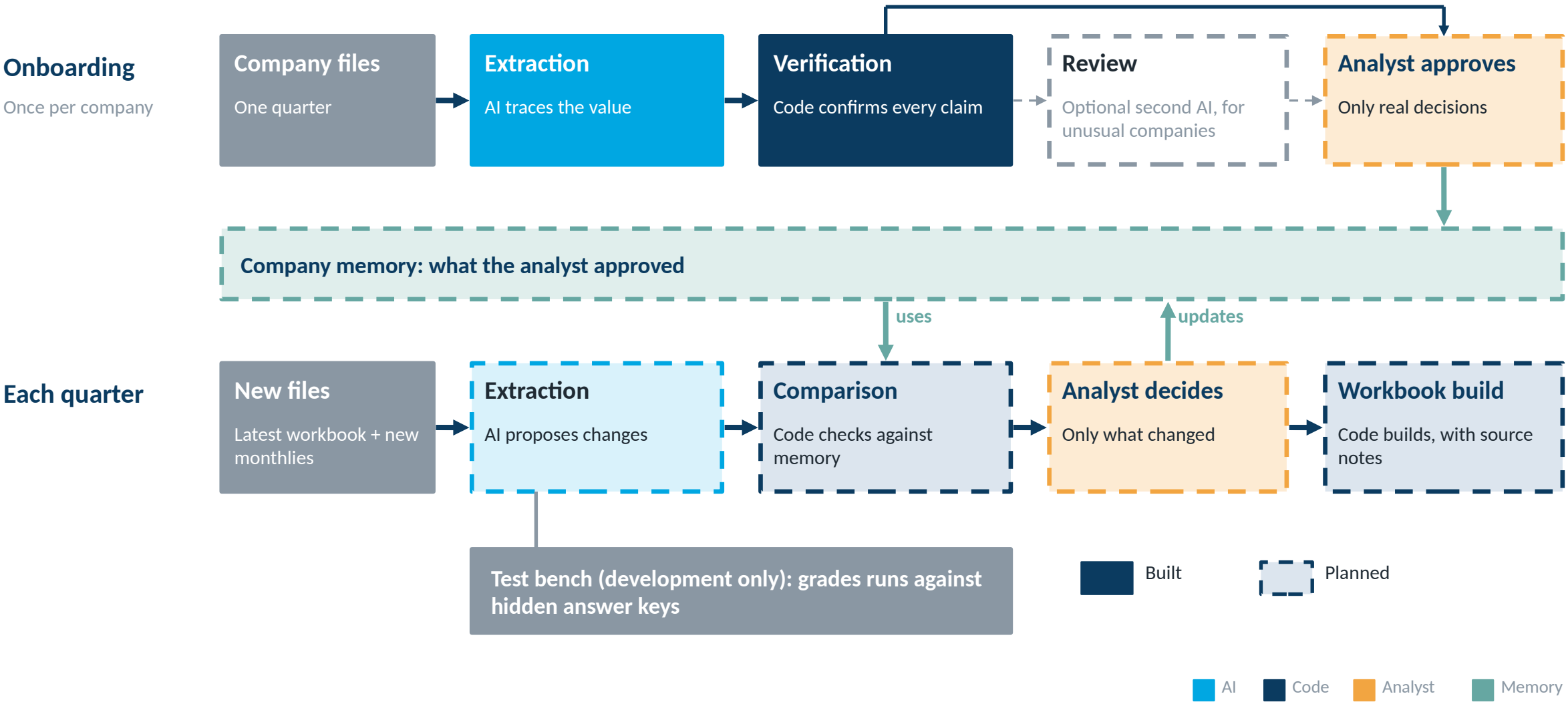
Safer. Slower, and the analyst finishes the workbook

Analyst AI suggestion

Also built: engagement letter approvals



Studio build (today) and the plan



How a Studio agent runs

Microsoft's harness: runs the AI and picks which tools to use

AI model

Claude Opus 5.5. Reads the files, reasons, proposes

Instructions

What the agent should do, in words

Knowledge base

XPV's valuation documents, read whole

Skills

Our tested code. The AI decides when to use it

Sandbox

Temporary workspace where code runs. No internet; wiped after each conversation

Tools and connectors

The only way out, e.g. saving to SharePoint

Hooks (preview)

Automatic checkpoints, e.g. before the AI runs code

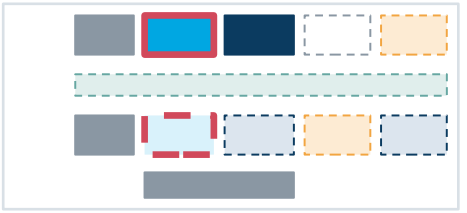
Where the platform has gaps, we add our own guard

Platform gap	Our guard
The AI can choose not to use a skill	Required steps started by a flow, not by the AI
Hooks are a test version and fail open	Final gate: no checking-code report, no result
Sandbox is temporary; nothing is kept	Memory and outputs saved to SharePoint
The AI can edit Excel files directly	Workbook changes only through our tested code

Our final check doesn't depend on Microsoft's.

AI Code Problem Working

Extraction: tracing the valuation



You are here

1 Identify each file and the current period

2 Start at the fund value and trace back to the monthly numbers

3 Confirm each link forward to the fund

4 Flag look-alikes and questions

Labels can mislead

"Adj. EBITDA" might really mean last-twelve-months earnings, adjusted, on a pro forma basis.

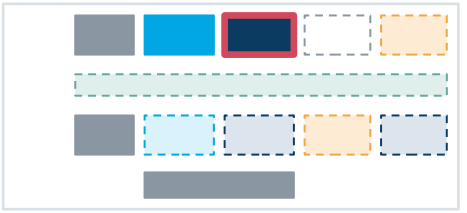
The company value cell multiplies the multiple by one specific cell. Following that formula shows which earnings figure is used.

Look-alikes it had to tell apart

- Last quarter's earnings next to this quarter's
- Earnings before and after adjustments
- A sheet named for an old period that holds current numbers

Onboarding mode is built. Quarterly mode (only what changed) is planned.

Verification: how code confirms the AI's work



You are here

AI claims

Cell F12 says "TTM Revenue"

It holds 18,400,000

Its formula adds 12 monthly cells

Those months are July 2025 to June 2026

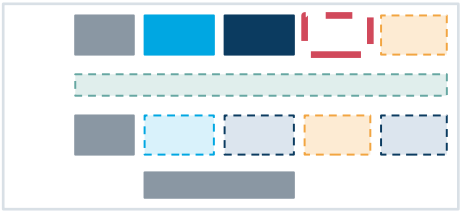
Code opens the file and confirms

- ✓ F12 really says "TTM Revenue"
- ✓ The value matches
- ✓ The formula points to those 12 cells
- ✓ The headers say July to June
- ✓ The 12 numbers add up to 18,400,000

Code proves the AI read the file correctly. The analyst decides whether it's the right number to use.

AI Code

Independent review (optional)



You are here

9 of 9

core answers agreed

Strong enough without it

Differences between the two (mostly wording)

38

Questions sent to the analyst

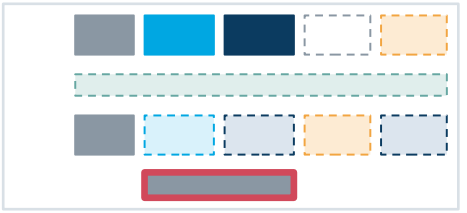
34

Target: only real decisions

about 5

Dashed = target

Testing and answer keys



You are here

253

tests passing

Code tests: invented workbooks with planted mistakes, all caught

Hidden

answer keys

Written by a person before any run; never shown to the AI. Analyst review planned

Same files

every run

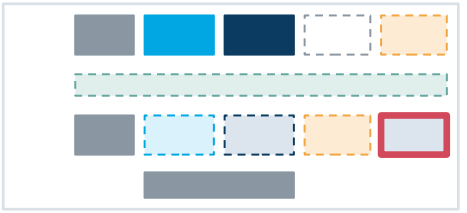
Each file gets a digital fingerprint when the AI reads it; grading only runs if the files are identical

Examples the tests catch

A pasted number claimed as a formula link · 11 months added up instead of 12 · last quarter's numbers used as a decoy · company value forced to match the fund · hidden sheets · a different set of files in the same run

Every run: AI ⇄ code checks, until every claim is proven or sent to the analyst. **Development only:** test bench grades → I improve the skills → run again.

Workbook build (planned)



You are here

Inputs

Latest workbook + new monthlies

AI proposes

Each change: value, source cell, target cell

Code checks

Each proposal, against files and memory

Code sorts

Pass, to the analyst, or stop

Code changes

Standard actions only, with a source note on every typed number

Code checks all

Whole workbook, before vs after. Out: new workbook + change log

Not confident? To the analyst

Open question: fill in a highlighted suggestion, or leave it blank?

Confidence: only when every check passes, it matches memory exactly, no sources disagree, and the change from last quarter is in a normal range. Code decides this, not the AI.

Change log: "this quarter only" (a one-off adjustment) or "permanent, saved to memory" (for example an acquisition, with new rows approved once).

AI Code Analyst

Caveats and known weaknesses

Caveats on today's results

- Two companies, one run each
- Answer keys written by me; analyst review planned
- 9 metrics is a small sample: one miss is 89%
- A third, differently built company is the real test
- One right period proven from numbers, not date headers
- 12 extra checks on Opus not yet graded

Known weaknesses, even in the target

- An approved mistake repeats every quarter
- Pasted numbers can be matched, not traced to a source
- New sheets or acquisitions always go to the analyst
- The analyst may wave items through
- The AI may skip its tools, and Microsoft's checkpoints can fail open

What we're doing about it

- A third, differently built company next
- Three repeat runs per company
- Analysts review the answer keys
- Quarter-on-quarter checks flag unusual changes
- Our own final gate: report plus fingerprints
- One design for every company; company details live only in memory

Problem Working

Glossary

Roll-forward	Updating last quarter's workbook with the new quarter's numbers
Studio build (today)	The Copilot Studio agents built so far: extractor, optional reviewer, test bench
Knowledge base	XPV's documentation of how valuations are built, including add-backs and categories
Trace	Follow formulas from one cell back to where its numbers come from
Answer key	Correct answers written before any run, hidden from the AI, used only to grade
Company memory	What the analyst approved for a company, reused each quarter
Backtest	A test on past quarters where we already know the right answer
Multiple	The number the analyst picks to turn earnings into a company value
Look-alike number	A real number that seems right but isn't the one to use
Investor report	What XPV reports to its investors

Glossary (continued)

Studio build	The streamlined target: extractor, checking code, workbook builder, company memory
Extractor	The AI agent that traces how a company's value is built
Test bench	Grades runs against hidden answer keys; development only
Harness	Microsoft's engine that runs the AI and picks which tools it uses
Skill	A packaged set of instructions and tested code the AI can use
Sandbox	A temporary workspace where code runs; wiped after each conversation
Hook (preview)	An automatic checkpoint the platform runs at set moments; still a test version
Fails open / fails closed	If a check breaks: "open" lets things through, "closed" stops them
Fingerprint	A digital code for a file; any change to the file changes the code
Connector	A configured link to another system, such as SharePoint

Sources

Vals AI, Excel Modeling Benchmark (updated October 12, 2026)

Best AI scores on Excel finance models; formulas vs numbers; long chains score lowest

Microsoft, GitHub Copilot harness, agent skills and richer context (2026)

What the harness is; skills; each task runs in a sandbox

Microsoft Copilot Studio team, The New Copilot Studio Agent Sandbox (2026)

The AI decides whether to use a skill; no internet; nothing kept

Microsoft Copilot Studio team, Hooks in Copilot Studio (October 12, 2026)

Automatic checkpoints; preview; fail open, so essential checks must be enforced separately

OpenAI, Why language models hallucinate (September 2025)

AI is rewarded for guessing over admitting doubt

METR, Recent frontier models are reward hacking (June 2025)

Top models take shortcuts while aware it isn't what users want

Cursor, Reward hacking is swamping model intelligence gains (June 2026)

Shortcuts are more common in newer models